

UNIVERSITY OF ASIA PACIFIC
DEPARTMENT OF CSE

COURSE CODE: CSE 402
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EXPERIMENT: Lab Report 7 - FCFS & SSTF Disk Scheduling

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Lab Report: FCFS & SSTF Disk Scheduling

Problem Statement

Implement FCFS (First Come First Serve) and SSTF (Shortest Seek Time First) disk scheduling algorithms for the given disk request queue. Determine the order in which the requests are served, calculate the total seek time for both algorithms, and compare their performance to identify which algorithm is more efficient for the given data.

Objective

The objective of this experiment is to understand how FCFS and SSTF disk scheduling algorithms move the disk head to serve pending requests. The lab also compares the total head movement (seek time) produced by the two methods and explains why one method performs better for the given request sequence.

Given Data

Item	Value
Request Queue	[0, 30, 41, 62, 14]
Initial Head Position	20

Working Principle

FCFS Disk Scheduling: FCFS serves disk requests strictly in the order in which they appear in the request queue. The disk head moves from the current position to the first request, then to the second request, and continues in the same sequence until all requests are completed.

SSTF Disk Scheduling: SSTF selects the pending request that is closest to the current disk-head position. After serving that request, the algorithm again searches the remaining requests and chooses the nearest one. This process continues until the queue becomes empty.

Seek Time Calculation

Seek time or head movement is calculated using the absolute distance between the current head position and the next selected request: |Current Position - Next Request|.

FCFS Calculation

Movement	Seek Distance
20 -> 0	20
0 -> 30	30
30 -> 41	11
41 -> 62	21
62 -> 14	48
Total	**130**

FCFS Order: [0, 30, 41, 62, 14] FCFS
Total Seek Time = 130

SSTF Calculation

Movement	Seek Distance
20 -> 14	6
14 -> 0	14
0 -> 30	30
30 -> 41	11
Movement	**Seek Distance**
41 -> 62	21
Total	**82**

SSTF Order: [14, 0, 30, 41, 62]

SSTF Total Seek Time = 82

Advantages and Disadvantages of FCFS

Advantages:

- Simple and easy to understand and implement.
- Requests are served fairly according to their queue order.
- No starvation occurs because every request eventually gets service.

Disadvantages:

- The disk head may travel a long distance when consecutive requests are far apart.
- It can produce high total seek time and lower disk performance.
- It does not consider the current head position when selecting the next request.

Advantages and Disadvantages of SSTF

Advantages:

- Selects the nearest request, so unnecessary disk-head movement is reduced.
- Usually provides lower average seek time than FCFS.
- Can improve disk access efficiency when requests are distributed across different tracks.

Disadvantages:

- Requests far from the current head can wait for a long time and may suffer starvation.
- The scheduler must repeatedly search for the nearest request, so it is more complex than FCFS.
- It is not as fair as FCFS because service order depends on distance rather than arrival order.

Why SSTF Is Better for This Experiment

For the given request queue, SSTF requires a total seek time of 82, while FCFS requires 130. SSTF reduces the total head movement by 48 units, which is about 36.9% less than FCFS. The reason is that SSTF always chooses the closest pending request to the current head position, avoiding several large unnecessary movements. Therefore, SSTF performs better for this specific set of requests.

Comparison between FCFS and SSTF

Point	FCFS	SSTF
Selection Rule	Follows request queue order	Chooses nearest pending request
Initial Head Position	20	20

Service Order	0, 30, 41, 62, 14	14, 0, 30, 41, 62
Total Seek Time	130	82
Starvation	No	Possible for distant requests
Fairness	High	Lower than FCFS
Implementation	Very simple	More decision making
Performance in This Lab	Higher seek time	Lower seek time
Better for Given Data	No	Yes

Source Code

```
req = [0, 30, 41, 62, 14]
head = 20

print("FCFS Disk
Scheduling")
total = 0
current = head

for r in req:
    total = total +
abs(current - r)
    current = r

print("Order:", req)
print("Total Seek Time:",
total)
print("\nSSTF Disk
Scheduling")

req2 = [0, 30, 41, 62,
14]
current = head
total = 0
order = []

while len(req2) > 0:
    near = req2[0]

    for r in req2:
        if abs(current -
r) < abs(current -
near):
            near = r
```

```
    total = total +
abs(current - near)
    order.append(near)
    current = near
    req2.remove(near)

print("Order:", order)
print("Total Seek Time:",
total)
print("\nSSTF is better
than FCFS")
```

Output

```
FCFS Disk Scheduling
Order: [0, 30, 41, 62, 14]
Total Seek Time: 130

SSTF Disk Scheduling
Order: [14, 0, 30, 41, 62]
Total Seek Time: 82

SSTF is better than FCFS
```

Discussion

FCFS and SSTF were tested using the same request queue and initial head position. FCFS followed the requests in their given order and resulted in 130 units of head movement. SSTF selected the nearest request each time, reducing the movement to 82 units. This shows that SSTF performed better for these requests. However, some distant requests may have to wait longer. FCFS is easier to implement and follows the original order, but it can result in more head movement.

Conclusion

In this lab, FCFS and SSTF disk scheduling algorithms were implemented and compared. SSTF performed better for the given requests, with 82 units of head movement compared to 130 for FCFS. Although SSTF reduced the total movement, FCFS is simpler and processes requests in their original order.

Github Link : CSE-402/lab 7 FCFS_SSTF Disk scheduling at main · 23101149-cmyk/CSE-402